

# Simran Dhankar

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## EDUCATION

- University of Washington** | *M.S. in Data Science – GPA: 3.9/4.0* | Seattle, WA Sep 2025 – Mar 2027
- Coursework: Machine Learning, Deep Learning, NLP, Agentic AI, Computer Vision, Statistics, Data Visualization, Linear Algebra
- GGSIIP University** | *B.Tech in Computer Science & Engineering – GPA: 3.8/4.0* | India Dec 2020 – Jun 2024
- Coursework: Object-Oriented Programming, Database Systems, Operating Systems, Data Structures & Algorithms

## EXPERIENCE

- Machine Learning Engineer Intern** | *Bajaj Capitals* | Gurugram, India Dec 2024 – Jun 2025
- Architected an end-to-end financial analytics ML pipeline ingesting 50K+ transaction records daily, applying feature engineering and unsupervised anomaly detection to lift structured data quality from 72% to 94%, directly reducing downstream model errors.
  - Built a regression-based retirement forecasting model incorporating 20+ behavioral and financial features, reducing RMSE by 18% over baseline and improving long-range prediction stability for customer portfolio planning.
  - Implemented k-fold cross-validation, residual error diagnostics, and SHAP-based feature attribution to harden model generalization, cutting overfitting gap from 11% to 4% on held-out validation sets.
  - Automated data ingestion and model retraining pipeline using Python and SQL, reducing manual intervention by 60% and enabling weekly model refresh cycles with zero downtime.
- Data Science Intern** | *Truminds Software Systems* | Gurugram, India Mar 2024 – Jul 2024
- Engineered 15+ geospatial features from 5K+ location records using spatial joins and distance-based aggregations, boosting downstream customer targeting model precision by 30% (0.61 to 0.79 precision@k), directly improving customer experience optimization.
  - Conducted comprehensive EDA including outlier detection, missing-value imputation, and geo-cluster analysis, surfacing 3 high-value customer micro-segments previously invisible in raw data, adopted directly in GTM strategy.
  - Integrated Google Maps API with a Flask backend to enrich customer profiles with real-time POI and footfall data, improving address-level geocoding accuracy by 20% and reducing lookup latency by 35%.
- Software Development Engineer Intern** | *ITMTB Technologies* | Pune, India Aug 2022 – Sep 2022
- Delivered a full-stack healthcare analytics platform (React, Node.js, Express, SQL, TypeScript) with JWT auth and role-based access control, deployed on AWS EC2/S3 and serving 500+ concurrent users.
  - Designed and optimized RESTful APIs with indexed SQL queries and Redis-layer caching, achieving a 40% reduction in average API response time (320ms to 192ms) under peak load.

## PROJECTS

- StudyMate AI – RAG-Based Student Productivity Agent** | *Python, Streamlit, FAISS, Groq API, LLMs* [Live Demo](#)
- Built a retrieval-augmented generation AI agent for document question answering, summarization, and personalized study planning using LLMs, improving response relevance by ~30% across 50+ test queries on multi-document inputs.
  - Designed an end-to-end NLP pipeline with text chunking, vector embeddings, and FAISS-based top-k retrieval, reducing hallucinated responses by ~25% and improving answer grounding through semantic search.
- MoLoRAG – Multi-Modal Logic-Aware RAG System** | *VLMs, PyTorch, HuggingFace, BERT, T5* [GitHub](#)
- Developed a multi-modal retrieval-augmented generation pipeline combining vision and text inputs, improving multi-hop QA performance (MRR +12%, Recall@5 +18%) on benchmark datasets.
  - Implemented advanced generative AI pipelines using LLMs, T5, and BERT for reasoning-aware text generation, outperforming GPT-3.5 zero-shot baselines by 8 F1 points on held-out test sets.
- Deepfake Detection System** | *Python, PyTorch, CNN, OpenCV, Streamlit* [Live Demo](#)
- Designed an end-to-end research prototype for deepfake detection processing 10K+ media samples via automated frame extraction, normalization, and spatial feature engineering on large-scale data.
  - Trained EfficientNet-B4 and custom CNN architectures achieving 90% validation accuracy vs. 73% ResNet-18 baseline (17-point gain), with early stopping and data augmentation preventing overfitting.
  - Deployed interactive Streamlit demo enabling real-time inference (<200ms/frame), generalized across 4 deepfake generation methods (Face2Face, DeepFaceLab, SimSwap, FSGAN).

## SKILLS

**ML & AI:** Statistical Modeling, Predictive Modeling, Generative AI (LLMs, VLMs, RAG), Transformers (BERT, T5), LangChain, LangGraph, Vector Databases (Pinecone, FAISS), Prompt Engineering, CNNs, RNNs, LSTMs, PyTorch, Scikit-learn, TensorFlow, Keras, XGBoost, AdaBoost, Random Forest

**Data & Analytics:** Pandas, NumPy, PySpark, Hypothesis Testing, A/B Testing, Experimental Design, ANOVA, PCA, Causal Inference, ROC-AUC, SHAP, Regression, Classification, Clustering, Data Visualization

**Languages & Frameworks:** Python, R, SQL, Java, JavaScript, TypeScript, React.js, Node.js, Express.js, MongoDB, Matlab

**Tools & Platforms:** AWS (EC2, S3), Databricks, HuggingFace, Git, Unix/Linux, Tableau, Matplotlib, Seaborn, Streamlit, OpenAI, Anthropic Claude

## ACHIEVEMENTS & LEADERSHIP

- 2nd Place, Databricks Hackathon (Nov 2025):** Built an ML model predicting flight connection disruption risk with 87% accuracy from 500K+ historical flight records, beat 30+ competing teams.
- Microsoft Learn Student Ambassador:** Designed and delivered technical workshops on Git/GitHub and Python fundamentals to 60+ students, increasing club workshop attendance by 40% semester-over-semester.
- Vice President, Enactus ADGITM:** Led a 40+ member cross-functional team across 3 social-impact projects, reaching 500+ community beneficiaries.